

MSDS 696 DATA SCIENCE PRACTICUM II | WEEK 6

Predicting U.S. Bank Distress

A proposal for ranking which banks need examining soonest

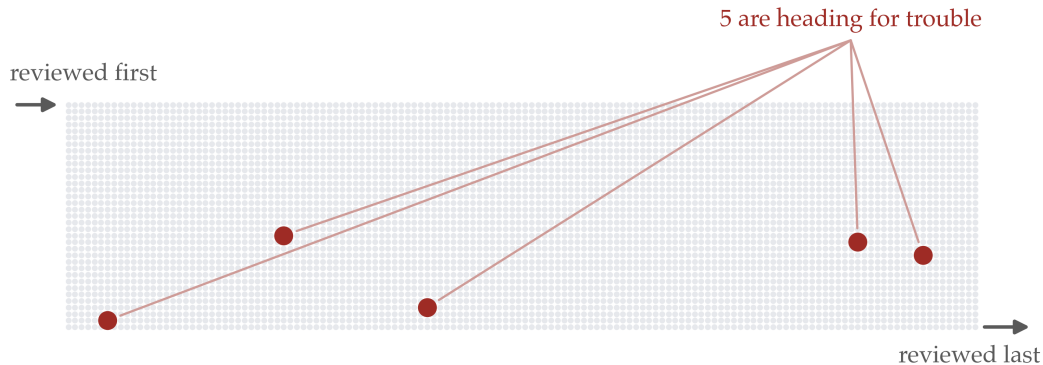
Oussama Ennaciri | August 2026

Supervisors should rank the review queue with a model instead of ranking on capital

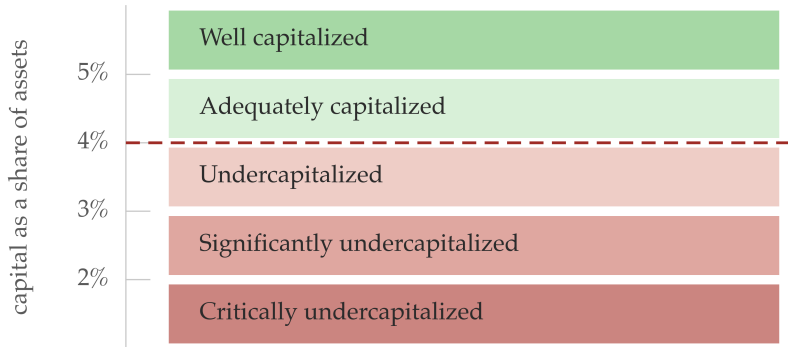
30% more

troubled banks caught than capital, for the same workload

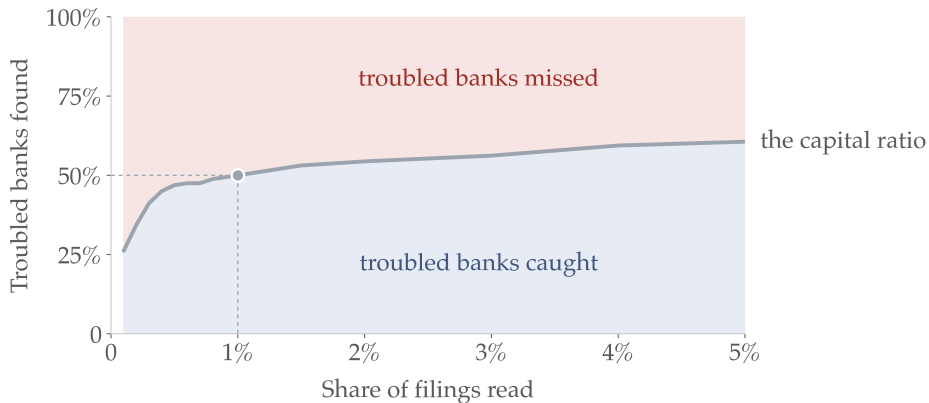
$\approx 4,900$ banks file every quarter, so the review queue has to put the potentially troubled ones first



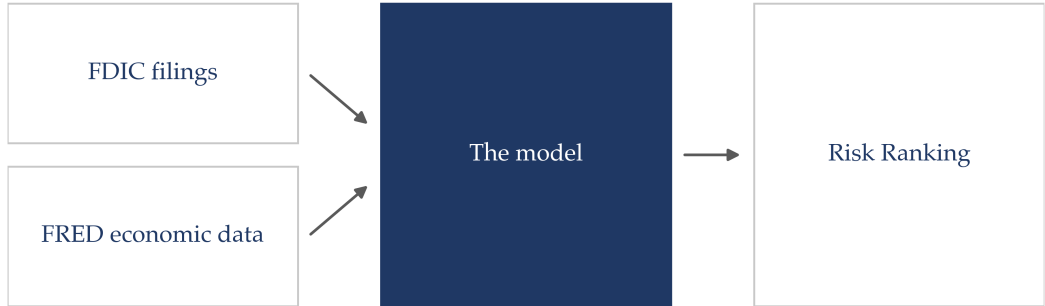
The obvious way to rank that queue is by capital against assets, the measure the regulation itself uses



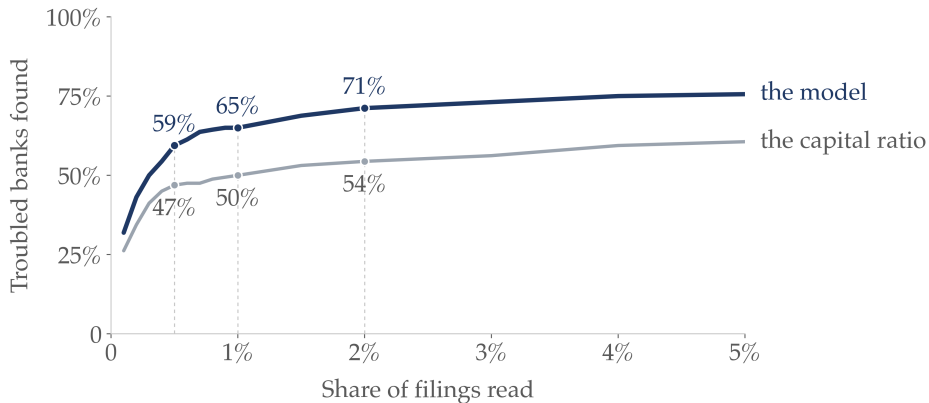
Ranked by capital ratio, half the troubled banks never reach the top of the list



An alternative is to rank the queue with a machine learning model trained on past filings



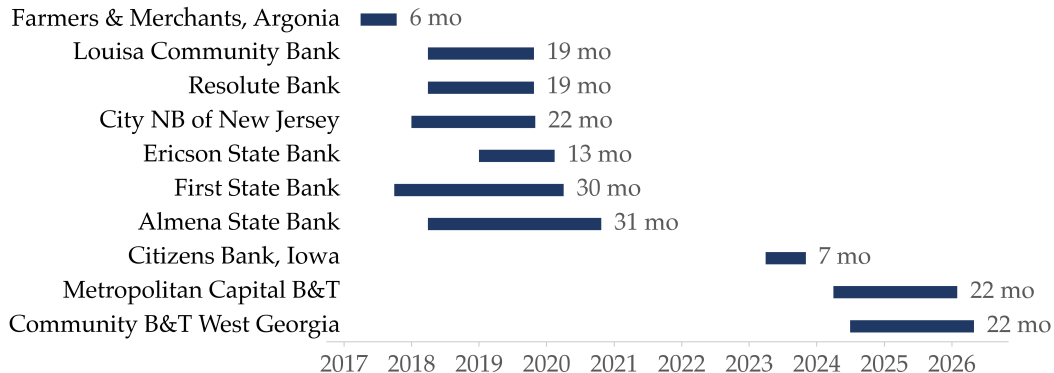
The model ranks the queue better than the capital ratio



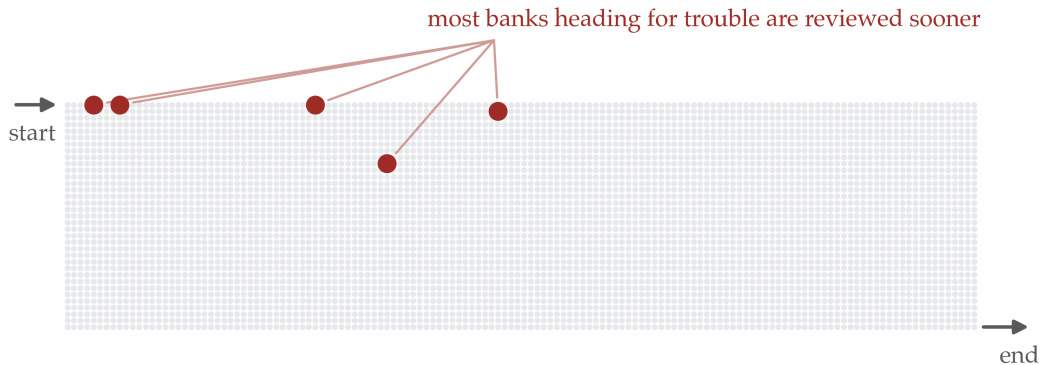
≈60% of the banks the model caught got a full year of warning



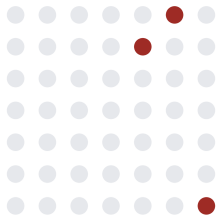
Every troubled bank that went on to fail was flagged first



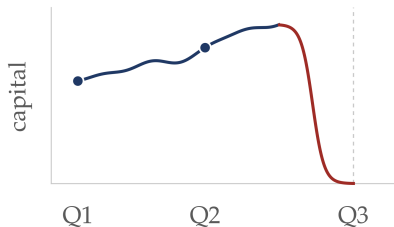
Rank the review queue with the model, not the capital ratio, and find 30% more troubled banks



Two limits come with both rankings



Most flags are false alarms



Cannot detect sudden bank runs